CORVALLE IMAGING SERVICES · LEARNING NEEDS ANALYSIS

Field Service Safety: Needs Analysis and Action Map

Prepared by Lisa Ferns, Instructional Designer · Scope: energy control and safety interlocks (Module 1 of Critical Decisions)

BUSINESS GOAL

Reduce safety near-misses and repeat service calls caused by skipped energy control steps by 30% within 12 months of launch.

What is happening

- Technicians pass the legacy safety course, but near-miss reports show a console shutdown used in place of lockout.
- Audits found E-stop circuits jumpered to keep systems running between visits.
- Repeat calls follow fuse replacements made without finding the cause.
- Most incidents happen under time pressure from the site.

How we found out

- 12 months of near-miss and incident reports
- Internal safety audit findings
- Interviews with 6 technicians and 3 site imaging managers
- 2 ride-alongs on service calls
- Review of the legacy course and its assessment

Why it is happening, and whether training can fix it

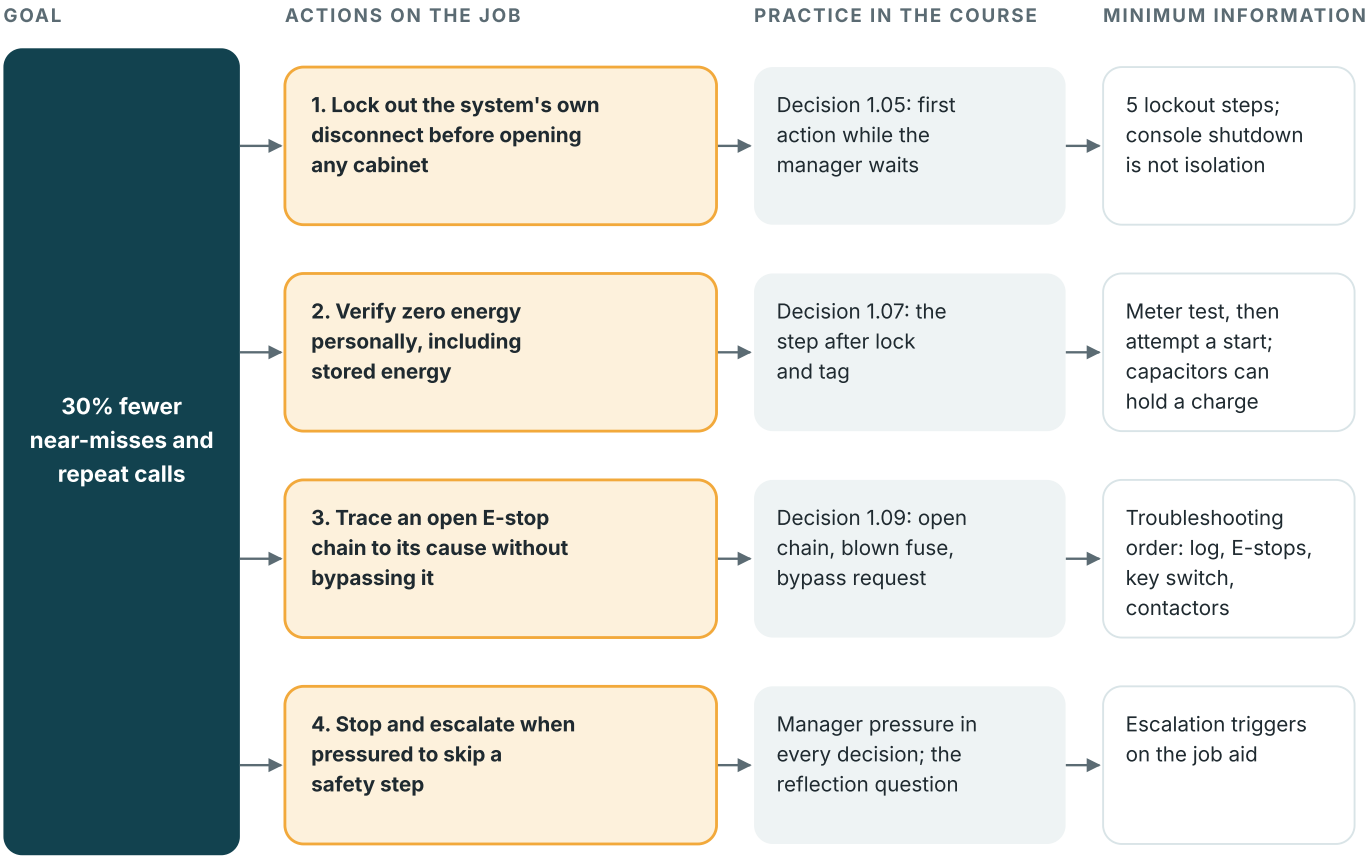
Training is only part of the answer. Two of the five causes are fixed by operations, not a course.

CAUSE	TYPE	TRAINING FIX?	RESPONSE
Technicians know the steps but skip them when a manager is waiting	Skill under pressure	Yes	Branching scenarios where pressure is part of every decision
No clear troubleshooting order for a system that won't stay on	Knowledge	Yes	Module 1 decision on the E-stop chain, plus the job aid
The legacy course tests recall, not decisions	Course design	Yes	Objectives rewritten at Apply and Analyze, with a capstone assessment
Site managers don't know how long a proper lockout takes	Environment	Partly	Scenarios practice saying no; a one-page briefing for site managers (operations)
Some service vans carry incomplete lockout kits	Environment	No	Standard kit checklist and quarterly van checks (operations)

All names, data and findings are fictional, created as a portfolio sample.

Action map: what technicians must do, and how they practice it

Each action is something a technician does on the job. Each one gets a realistic practice decision, and only the information needed to make it.



Recommendations

Training

A 7 to 10 minute branching scenario (Module 1) built in Storyline 360, with consequence-based feedback, a scored safety meter and a reflection question.

Performance support

A one-page job aid for every service van and tablet, matching the course: lockout steps, troubleshooting order, never list and escalation triggers.

Operations (non-training)

A site manager briefing on lockout time, and a standard lockout kit checklist with quarterly van checks.

How success will be measured

- **Level 1:** technician relevance and confidence survey at the end of the module.
- **Level 2:** safe choices on the first attempt at each decision, from the course data.
- **Level 3:** supervisor ride-along checklist at 30 and 90 days.
- **Level 4:** near-misses and repeat calls tied to energy control, reviewed quarterly against the 30% goal.